UE26CY151A · ENGINEERING CHEMISTRY

SEMESTER 1 · UNIT 1

03

FIGURES

Graphs & Diagrams

Redrawn energy-level diagrams, spectra, potential curves and phase diagrams, each with what to label in an exam.

M1
Molecular Spectroscopy
8 topics

M2
Phase Equilibria
5 topics

M3
Computational Chemistry
1 topic

Exam weight	★★★ high · ★★☆ medium · ★☆☆ low
Flags	NOT ON CURRENT SLIDES · TRANSCRIPT ONLY · ADDED FOR CLARITY
Sources	S004·8 slide 8 of deck 004 · T009 class transcript · PDF Unit-1 slide PDF · ESA past paper

Compiled from PESU AV-summary transcripts, class slides and past ESA papers · 13 October 2026

Contents

M1 Molecular Spectroscopy

T1 EM radiation & interaction with matter ★★★

T2 Electromagnetic spectrum & regions ★★★

T3 Born–Oppenheimer approximation ★★☆

T3b Beer–Lambert law ★☆☆

T4 Rotational (microwave) spectroscopy ★★★

T5 Vibrational (IR) – SHO ★★★

T6 Vibrational – anharmonic ★★☆

T7 Electronic spectroscopy & Franck–Condon ★★★

M2 Phase Equilibria

T8 Basic terms & phase rule ★★★

T9 Derivation of phase rule ★★★

T10 One-component system: water ★★★

T11 Two-component: reduced phase rule & Pb–Ag ★★★

T12 Thermal analysis / cooling curves ★★☆

M3 Computational Chemistry

T13 Computational chemistry ★☆☆

MODULE 1

Molecular Spectroscopy

T1 · EM radiation & interaction with matter | T2 · Electromagnetic spectrum & regions | T3 · Born–Oppenheimer approximation | T3b · Beer–Lambert law | T4 · Rotational (microwave) spectroscopy | T5 · Vibrational (IR) – SHO | T6 · Vibrational – anharmonic | T7 · Electronic spectroscopy & Franck–Condon

T1 EM radiation & interaction with matter ★★★

No graph for this topic.

T2 Electromagnetic spectrum & regions ★★★

No graph for this topic.

T3 Born–Oppenheimer approximation ★★☆

No graph for this topic.

T3b Beer–Lambert law ★☆☆

No graph for this topic.

T4 Rotational (microwave) spectroscopy

★★★

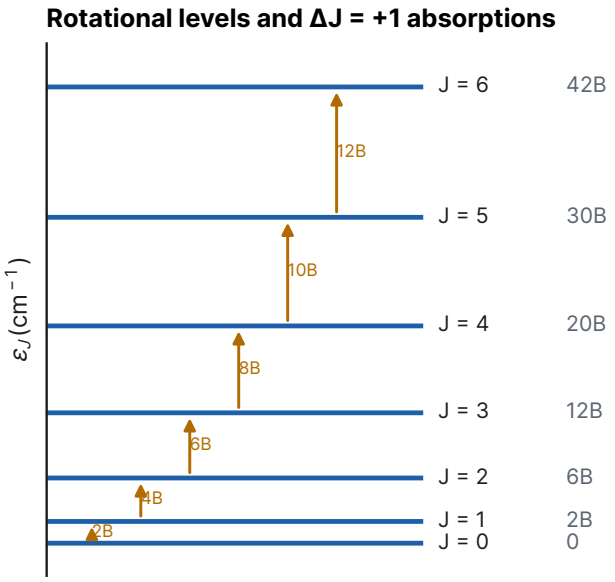


Fig. G9 Rigid-rotor energy levels J=0–6 (values J(J+1)B), ΔJ=+1 arrows.

IN THE EXAM Label J, the energy in B, and the widening gaps 2B,4B,6B...

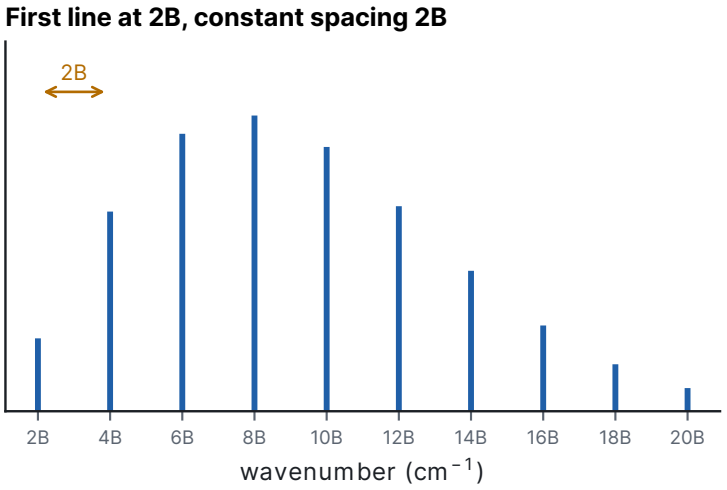


Fig. G10 Rotational stick spectrum.

IN THE EXAM Lines at 2B, 4B, 6B...; constant spacing 2B; intensity rises then falls.

T5 Vibrational (IR) – SHO

★★★

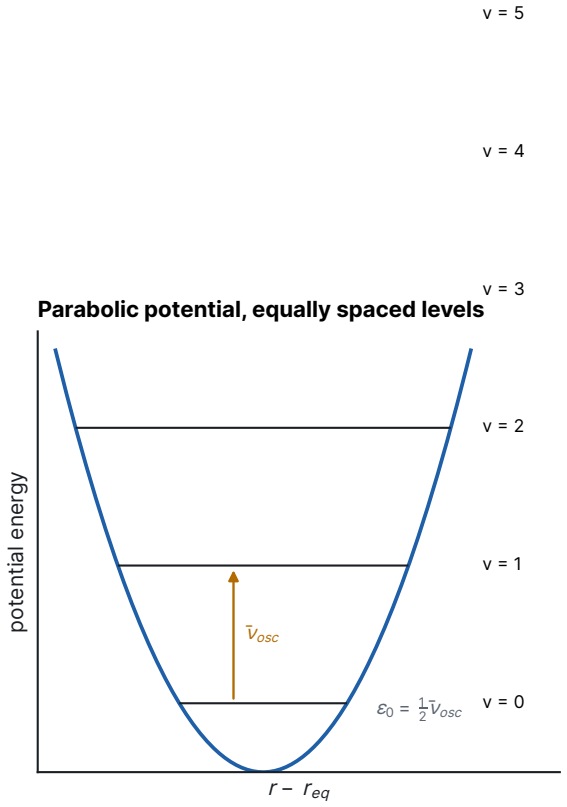


Fig. G13 SHO parabola with equally spaced levels.

IN THE EXAM Mark v=0…5 and $\varepsilon_0 = \frac{1}{2}\bar{\nu}$; spectrum is a single line.

T6 Vibrational – anharmonic

★★☆

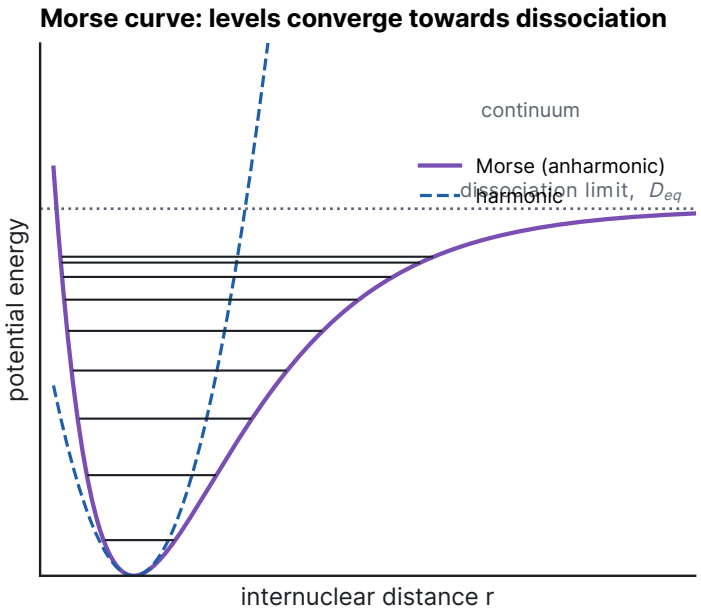


Fig. G15 Morse curve vs harmonic parabola.

IN THE EXAM Show D_{eq}, convergence of levels and dissociation limit.

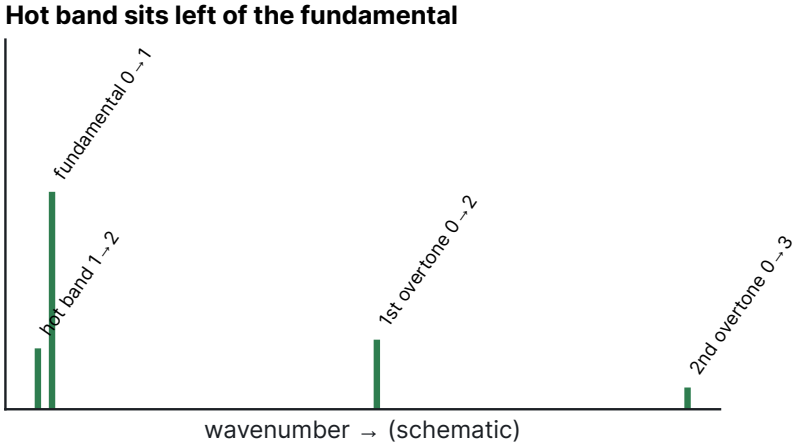


Fig. G17 Anharmonic IR spectrum (schematic).

IN THE EXAM Hot band left of fundamental; overtones with decreasing spacing.

T7 Electronic spectroscopy & Franck–Condon

★★★

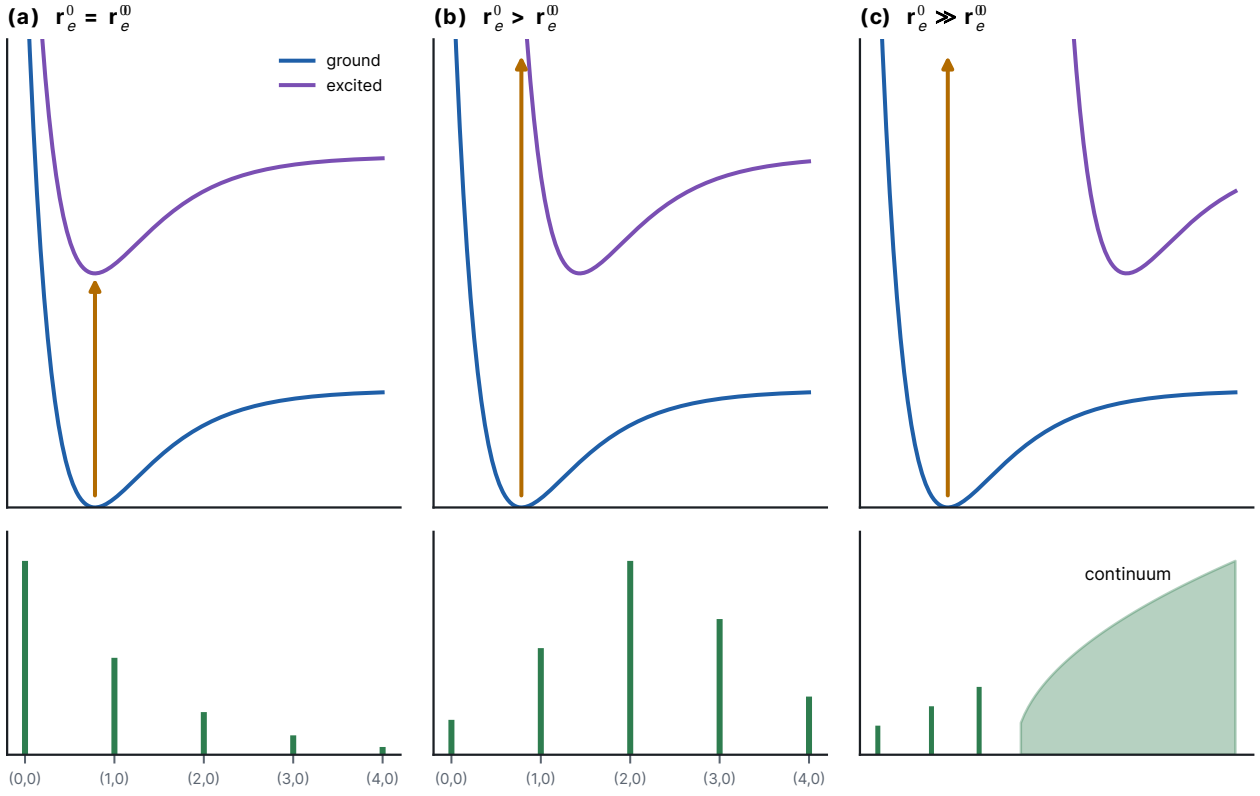


Fig. G21 Franck–Condon, three cases (a, b, c) with spectra.

IN THE EXAM Draw vertical arrow from v″=0; (c) ends in a continuum.

MODULE 2

Phase Equilibria

T8 · Basic terms & phase rule | T9 · Derivation of phase rule | T10 · One-component system: water | T11 · Two-component: reduced phase rule & Pb–Ag | T12 · Thermal analysis / cooling curves

T8 Basic terms & phase rule ★★★

No graph for this topic.

T9 Derivation of phase rule ★★★

No graph for this topic.

T10 One-component system: water ★★★

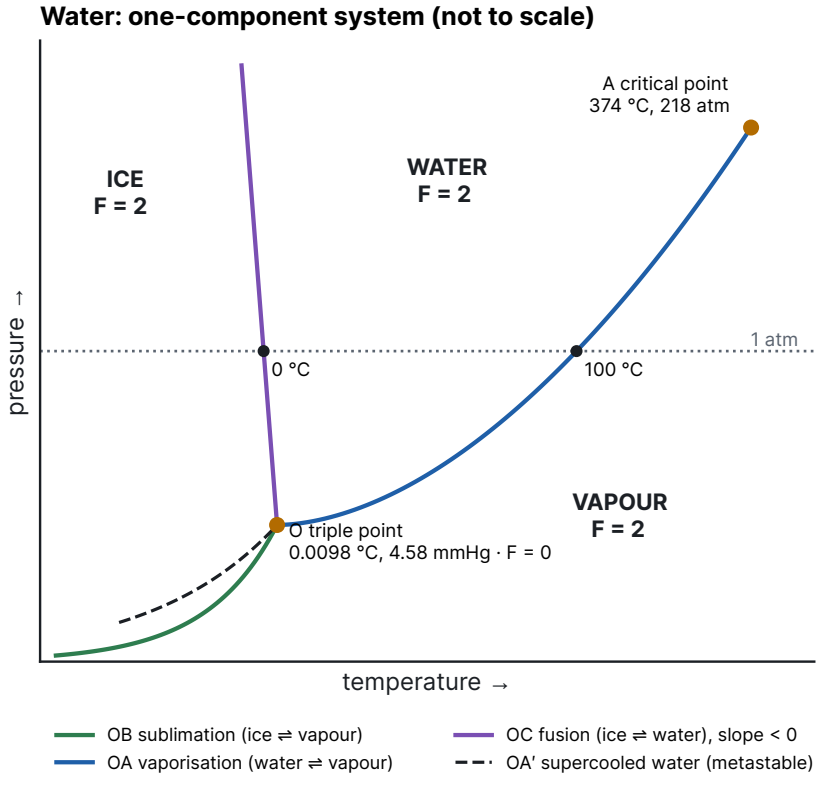


Fig. G22 Water phase diagram (schematic, not to scale).

IN THE EXAM Label OA, OB, OC, OA′, O, A, with T,P values and F in each region.

T11 Two-component: reduced phase rule & Pb–Ag ★★★

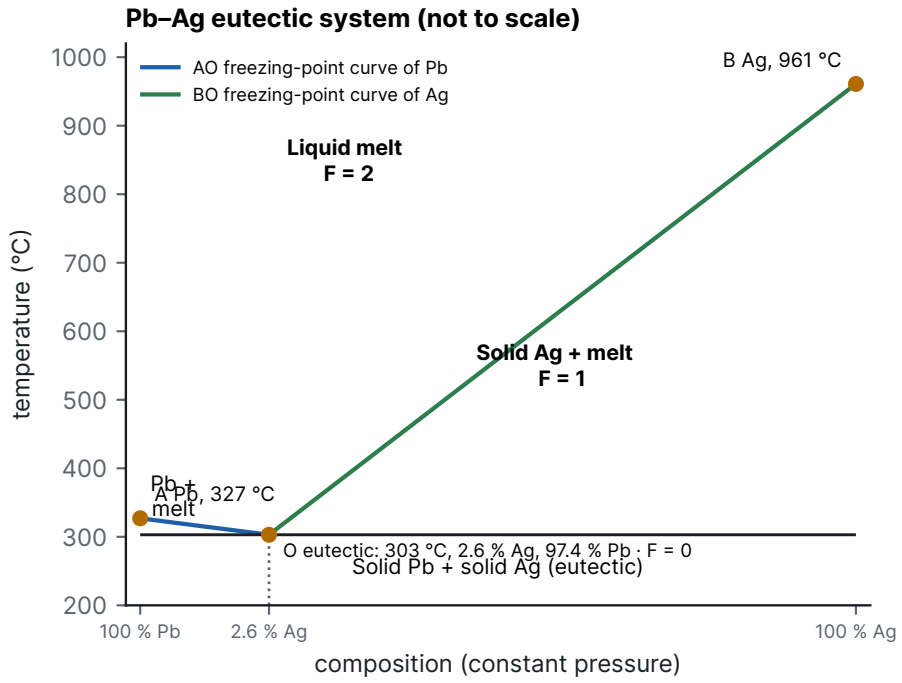


Fig. G23 Pb–Ag eutectic diagram (schematic; Pb left, Ag right).

IN THE EXAM Eutectic O at 303 °C, 2.6 % Ag; label the four areas. Slide lettering is inconsistent (see conflicts) – follow your faculty's.

T12 Thermal analysis / cooling curves ★★☆



Fig. G24 Cooling curves: pure, mixture, eutectic.

IN THE EXAM Mark break and halt.

MODULE 3

Computational Chemistry

T13 · Computational chemistry

T13 Computational chemistry

★☆☆

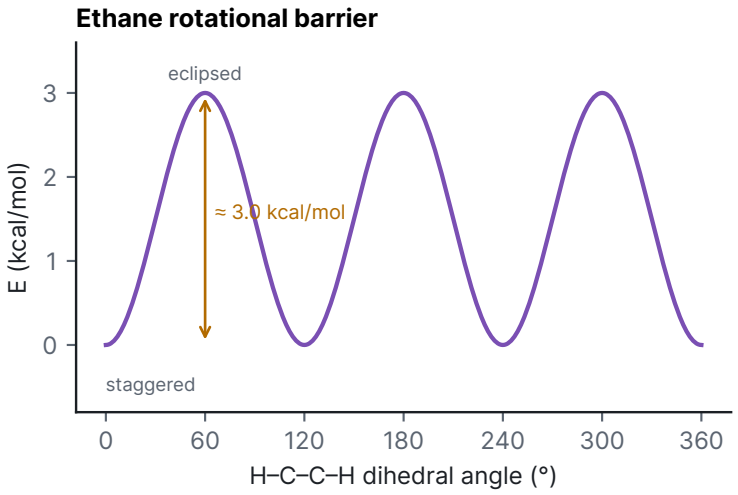


Fig. G25 Ethane torsion barrier ≈3.0 kcal/mol (illustrative).

IN THE EXAM Staggered minima, eclipsed maxima.